Today's poll: Pen Pencil

Ch 3.1: Linear Regression

Lecture 4 - CMSE 381

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Fri, Jan 21, 2022

Announcements

Last time:

- 2.2 Assessing Model Accuracy

Announcements:

- Get on slack!
  - +1 point on the first homework if you post a gif in the thread
- First homework due today! Covers:
  - Weds 1/12 lecture
  - Fri 1/14 Lab
  - Weds 1/19 lecture
- No lab this week
- Office hours

Covered in this lecture

- Least squares coefficient estimates for linear regression
- Residual sum of squares (RSS)
- Confidence interval, hypothesis test, and p-value for coefficient estimates
- Residual standard error (RSE)
- R squared

Section 1

Simple Linear Regression

Setup

quantitative => regression

- Predict Y on a single predictor variable X

quantitative

$$Y \approx \underbrace{\beta_0}_{} + \underbrace{\beta_1}_{} X$$

coefficients

- "$\approx$" .... "is approximately modeled as"

$y = mx + b$

intercept β_0

slope β_1

Example

Advertising

x y

		TV	Radio	Newspaper	Sales
1					
2	1	230.1	37.8	69.2	22.1
3	2	44.5	39.3	45.1	10.4
4	3	17.2	45.9	69.3	9.3
5	4	151.5	41.1	58.5	18.5
6	5	180.8	10.8	58.4	12.9
7	6	8.7	48.9	75	7.2
8	7	57.5	32.8	23.5	11.8
9	8	120.2	19.6	11.6	13.2
10	9	8.6	2.1	1	4.8
11	10	199.8	2.6	21.2	10.6
12	11	66.1	5.8	24.2	8.6

intercept slope

$$\text{Sales} \approx \beta_0 + \beta_1 \, TV$$

Coefficients / parameters

once we have good guesses for β_0, β_1

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

How do we estimate the coefficients?

Find β_0 and β_1 that minimize the least squares criterion!

definition of "good"

Given data $(x_1, y_1), \ldots, (x_n, y_n)$

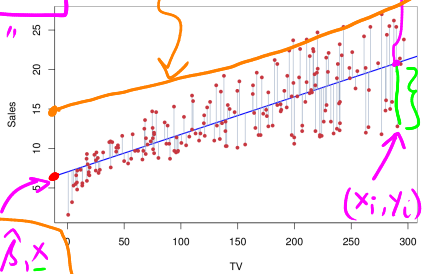
$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$

i^{th} residual

$e_i = y_i - \hat{y}_i$

line diff $\hat{\beta}_0$, same $\hat{\beta}_1$

$(x_i, \hat{y}_i)$

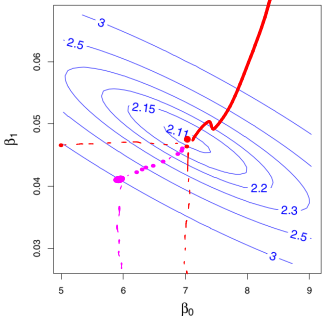


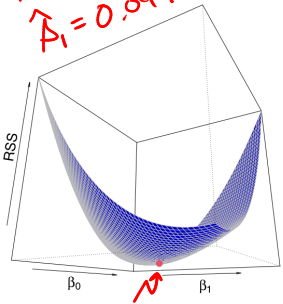
(x_i, y_i)

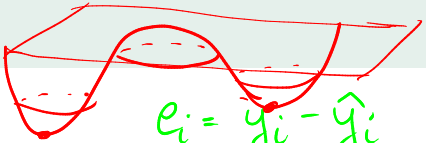
$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$

the line

RSS







$e_i = y_i - \hat{y}_i$

$\hookrightarrow \hat{\beta}_0 + \hat{\beta}_1 x_i$

Residual sum of squares RSS is

$$RSS = e_1^2 + \cdots + e_n^2$$
$$= \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

based on $\hat{\beta}_0$ and $\hat{\beta}_1$

$\hat{\beta}_0 = 6$

$\hat{\beta}_1 = 0.043$

$$\text{sales} \approx \beta_0 + \beta_1 \text{TV}$$

RSS = 3

Least squares coefficient estimates

$y_i - (\quad)$

$$RSS = \min_{\beta_0,\beta_1} \sum_i (\hat{\beta}_0 + \hat{\beta}_1 x_i)^2 \qquad \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$$

$\bar{x} = \frac{1}{n}\sum x_i$

$\bar{y} = \frac{1}{n}\sum y_i$

$$\frac{\partial RSS}{\partial \beta_0} = -2\sum_i (y_i - \beta_0 - \beta_1 x_i) = 0$$

$$\frac{\partial RSS}{\partial \beta_1} = -2\sum_i x_i(y_i - \beta_0 - \beta_1 x_i) = 0$$

$$\begin{cases} \hat{\beta}_1 = \dfrac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \\ \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \end{cases}$$

Closed form!

Section 2

Assessing Coefficient Estimate Accuracy

Bias in estimation

Analogy: sample mean

mean

- Assume a true value μ^*
- An estimate from training data $\hat{\mu}$
- The estimate is unbiased if $E(\hat{\mu} = \mu^*)$

random variable

$\frac{1}{n}\sum x_i$

- Sample mean is unbiased for population mean:

$$E(\hat{\mu}) = E\left(\frac{1}{n}\sum_i X_i\right) = \mu$$

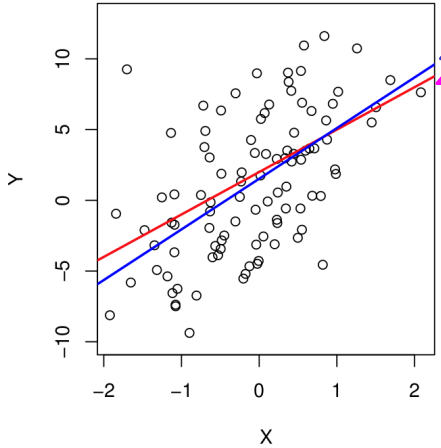
- Standard variance estimate is biased

$$E(\hat{\sigma}^2) = E\left[\frac{1}{n}\sum_i (X_i - \overline{X})^2\right] \neq \sigma^2$$

Linear regression is unbiased

red = true line

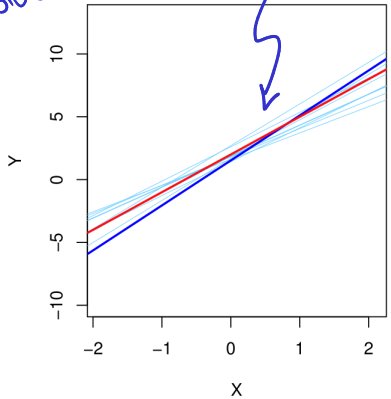
blue = $\hat{\beta}_0 + \hat{\beta}_1 X$



light lines

new $\hat{\beta}_0, \hat{\beta}_1$

lines for new samples



Variance in estimation

$Y = f(X) + \varepsilon$

noise

$\bar{X} = \frac{1}{n}\sum x_i$

- True value μ^*
- Estimate from training data $\hat{\mu}$
- Variance of sample mean $\mathrm{Var}(\hat{\mu}) = \mathrm{SE}(\hat{\mu})^2 = \frac{\sigma^2}{n}$

Standard error

- Variance of linear regression estimates:

$$\widehat{\mathrm{SE}(\hat{\beta}_0)}^2 = \hat{\sigma}^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}\right]$$

$$\widehat{\mathrm{SE}(\hat{\beta}_1)}^2 = \frac{\hat{\sigma}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

where $\sigma^2 = \mathrm{Var}(\varepsilon)$

- Estimate σ: $\hat{\sigma}^2 = \frac{RSS}{n-2}$

Residual sum of squares

Confidence Interval

(t-dist)

The 95% confidence interval for β_1 approximately takes the form

$$\hat{\beta}_1 \pm 2 \cdot \underbrace{\mathrm{SE}(\hat{\beta}_1)}$$

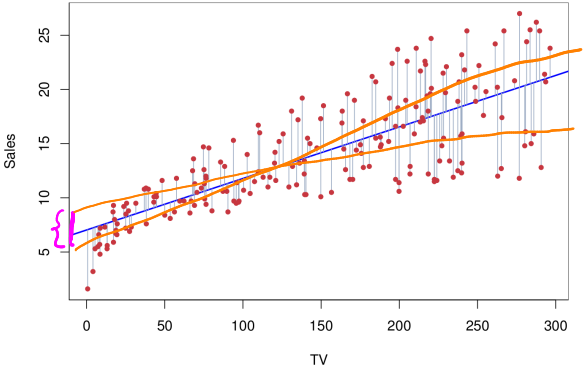
$$\left[\hat{\beta}_1 - 2\,\mathrm{SE}(\hat{\beta}_1),\ \hat{\beta}_1 + 2\,\mathrm{SE}(\hat{\beta}_1)\right]$$

Interpretation:
There is approximately a 95% chance that the interval

$$\left[\hat{\beta}_1 - 2 \cdot \mathrm{SE}(\hat{\beta}_1), \hat{\beta}_1 + 2 \cdot \mathrm{SE}(\hat{\beta}_1)\right]$$

will contain β_1 where we repeatedly approximate $\hat{\beta}_1$ using repeated samples.

CI in Advertising data



For the advertising data set, the 95% CIs are:

- β_1 :: $[0.042, 0.053]$
- β_0 : $[6.130, 7.935]$

Hypothesis testing

null hypothesis

H_0: There is no relationship between X and Y

H_A: There is some relationship between X and Y

alternative hypothesis

$H_0: \beta_1 = 0$

$$Y = \beta_0 + \beta_1 X$$
$$= \beta_0 + 0X$$
$$= \beta_0$$

$H_A: \beta_1 \neq 0$

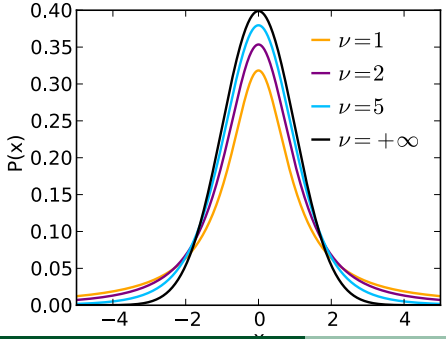
Test statistic and p-value

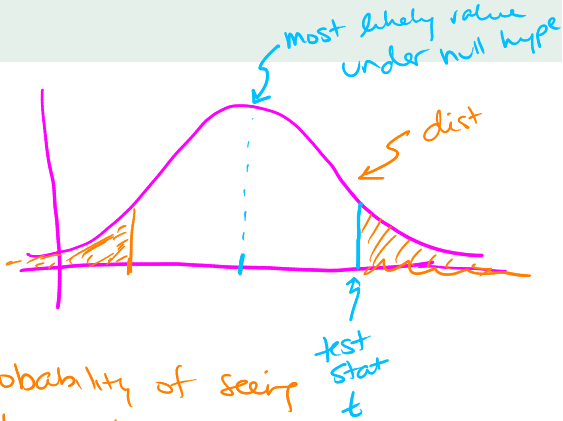
Test statistic:

$$\boxed{t = \frac{\hat{\beta}_1 - 0}{\mathrm{SE}(\hat{\beta}_1)}}$$

→ a number

t-distribution with $n - 2$ degrees of freedom





Probability of seeing a test stat more unlikely than my test stat = orange area

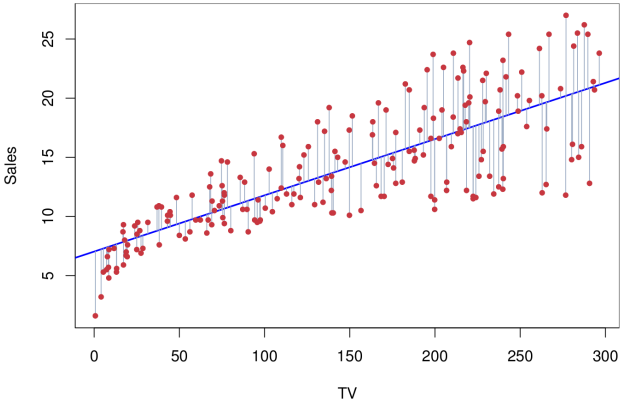
p-value =

Advertising example

$se(\hat{\beta}_i)$

Reject the null hypothesis

	Coefficient	Std. error	t-statistic	p-value
Intercept	$\hat{\beta}_0$ 7.0325	0.4578	15.36	< 0.0001
TV	$\hat{\beta}_1$ 0.0475	0.0027	17.67	< 0.0001



0.05

Assessing the accuracy of the module: RSE

Quantify the extent to which the model fits the data

Residual standard error (RSE):

$$RSE = \sqrt{\frac{1}{n-2} RSS}$$

$$= \sqrt{\frac{1}{n-2} \sum_i (y_i - \hat{y}_i)^2}$$

Better w/ more data

- Estimate of the std. dev. of ε
- Avg amount response will deviate from the true regression line

Assessing the accuracy of the module: R^2

amt of variability after regression

R^2 is proportion of variability in Y can be explained using X

R squared:

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

← small

where total sum of squares is

$$TSS = \sum_i (y_i - \bar{y})^2$$

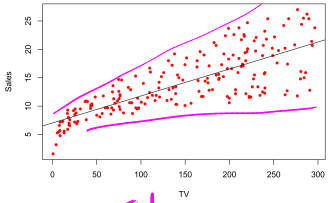
Always in [0, 1]

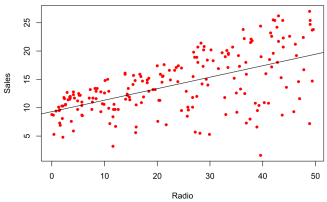
↳ total variance in response Y

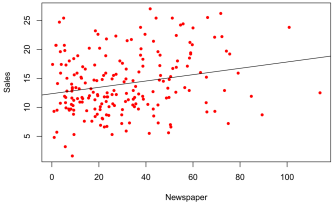
variability before regression

- Close to 1
  ↳ large proportion of variability is explained by the regression
- Close to 0
  ↳ regression does not explain much of the response variability

Advertising example







TV

Radio

Newspaper

$R^2 = 0.61$

$R^2 = 0.33$

$R^2 = 0.05$

Lots

Very little

How much do I trust the linearity assumption?

Correlation Coefficient

Measure dependence between two random variables X and Y

$$r = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)}\sqrt{\mathrm{Var}(Y)}}$$

$$= \frac{\sum_i (x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum_i (x_i - \overline{x})^2}\sqrt{\sum_i (y_i - \overline{y})^2}}$$

$\bar{x} = \frac{1}{n}\sum x_i$

$\bar{y} = \frac{1}{n}\sum y_i$

- Between -1 and 1

1: variables are perfectly related (same $(x = y)$)

0: variables are not linearly related

-1: Variables neg'ly related (different) (neg slope)

For linear regression $r^2 = R^2$

This won't generalize....

Correlation Coefficient

Measure dependence between two random variables X and Y

$$r = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)}\sqrt{\mathrm{Var}(Y)}}$$
$$= \frac{\sum_i (x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum_i (x_i - \overline{x})^2}\sqrt{\sum_i (y_i - \overline{y})^2}}$$

Next time

Today: 3.1

- Monday:
  - 3.2 Multiple linear regression
- Weds:
  - 3.3/3.5 Even more linear regression
- Fri:
  - Linear regression lab
  - HW #2 due

Today

Monday

Weds

For next year

- Ha, I just forgot slides and notes at the end. Needs cleanup!
- should include values on advert data to make sense
- RSE needs interpretation
- Not convinced by my R^2 vs r^2 discussion